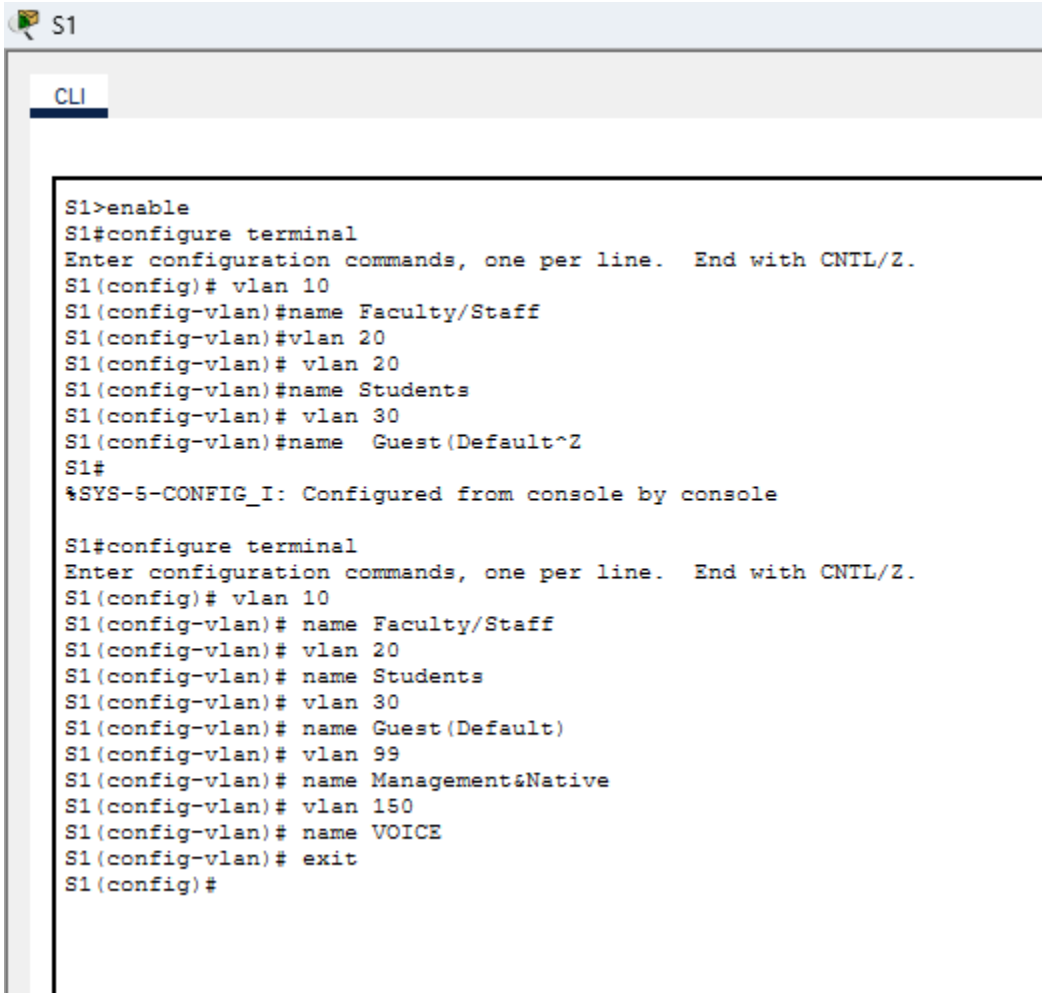
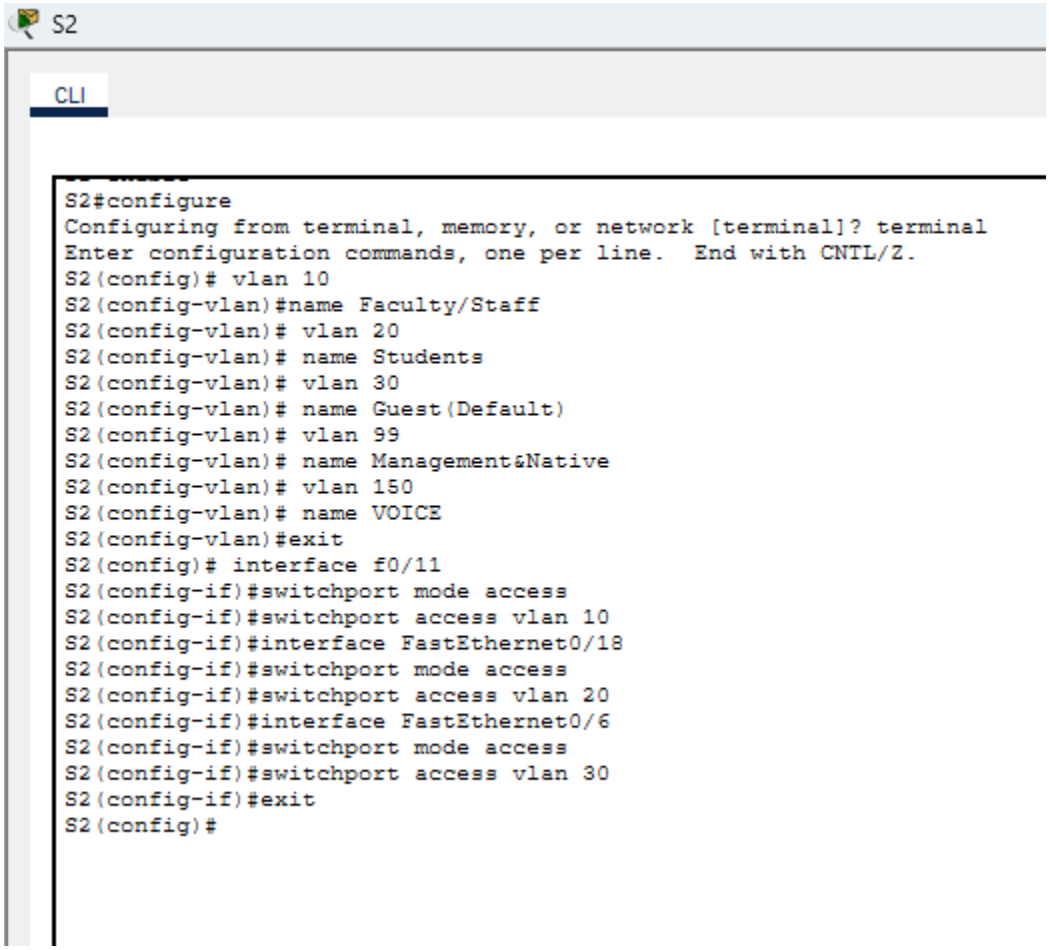
Lab3 - VLAN Configuration

1. Network Topology & Architecture
The network consists of three switches (S1, S2, S3) interconnected via trunk links, with end devices spread across different floors or locations but logically segmented into Virtual Local Area Networks (VLANs).
● S1 (Core/Distribution Switch): Acts as the central aggregation point connecting to S2 and S3.
● S2 (Access Switch): Hosts PC1, PC2, and PC3.
S3 (Access Switch): Hosts PC4, PC5, PC6, and a Cisco IP Phone on $F0/11$.

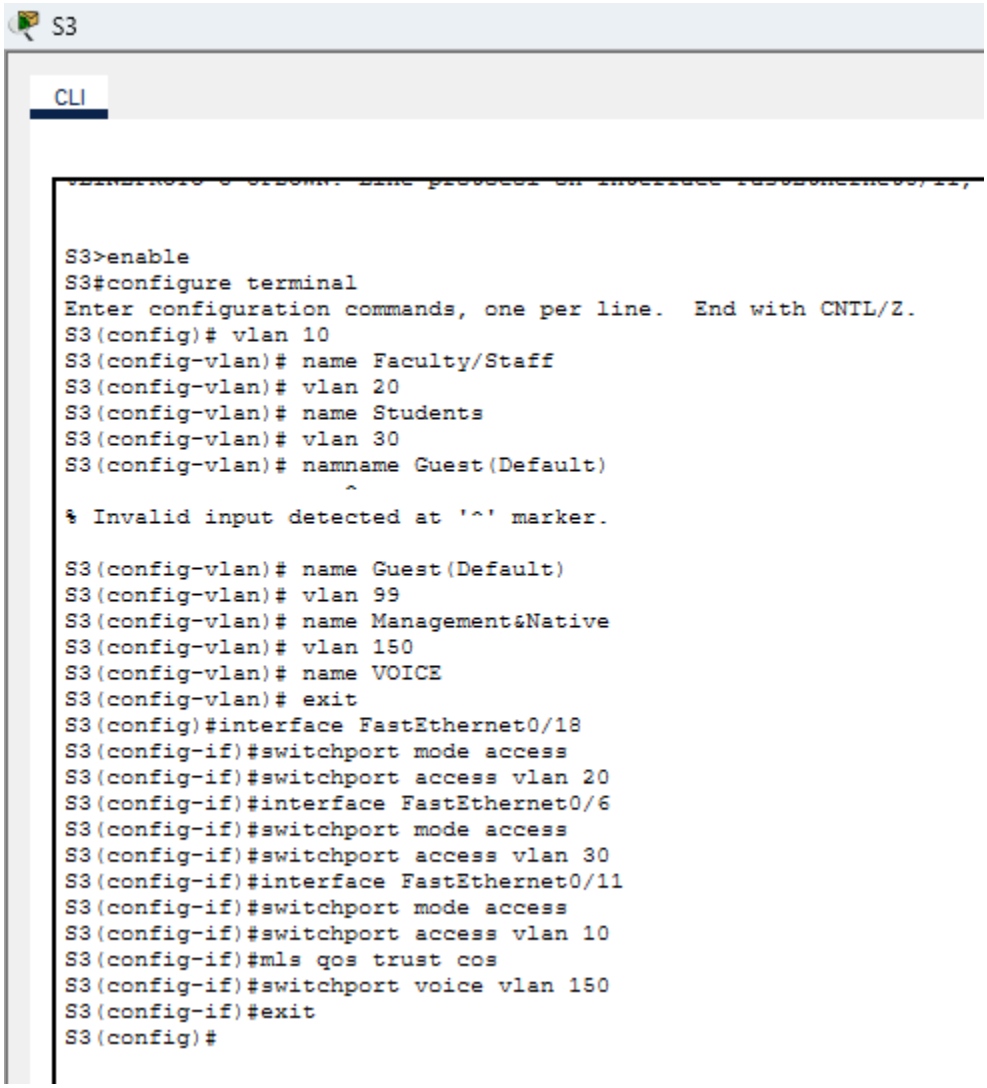
2. Complete Device Configurations
Switch 1 (S1) Configuration
S1 requires the creation of all global VLAN databases. Trunking commands are implied on its uplink interfaces to pass multi-VLAN traffic to S2 and S3.



Switch 2 (S2) Configuration
S2 handles local access ports for PC1, PC2, and PC3.



Switch 3 (S3) Configuration
S3 requires access port assignments and unique Quality of Service (QoS) and Voice VLAN mapping on interface $F0/11$.

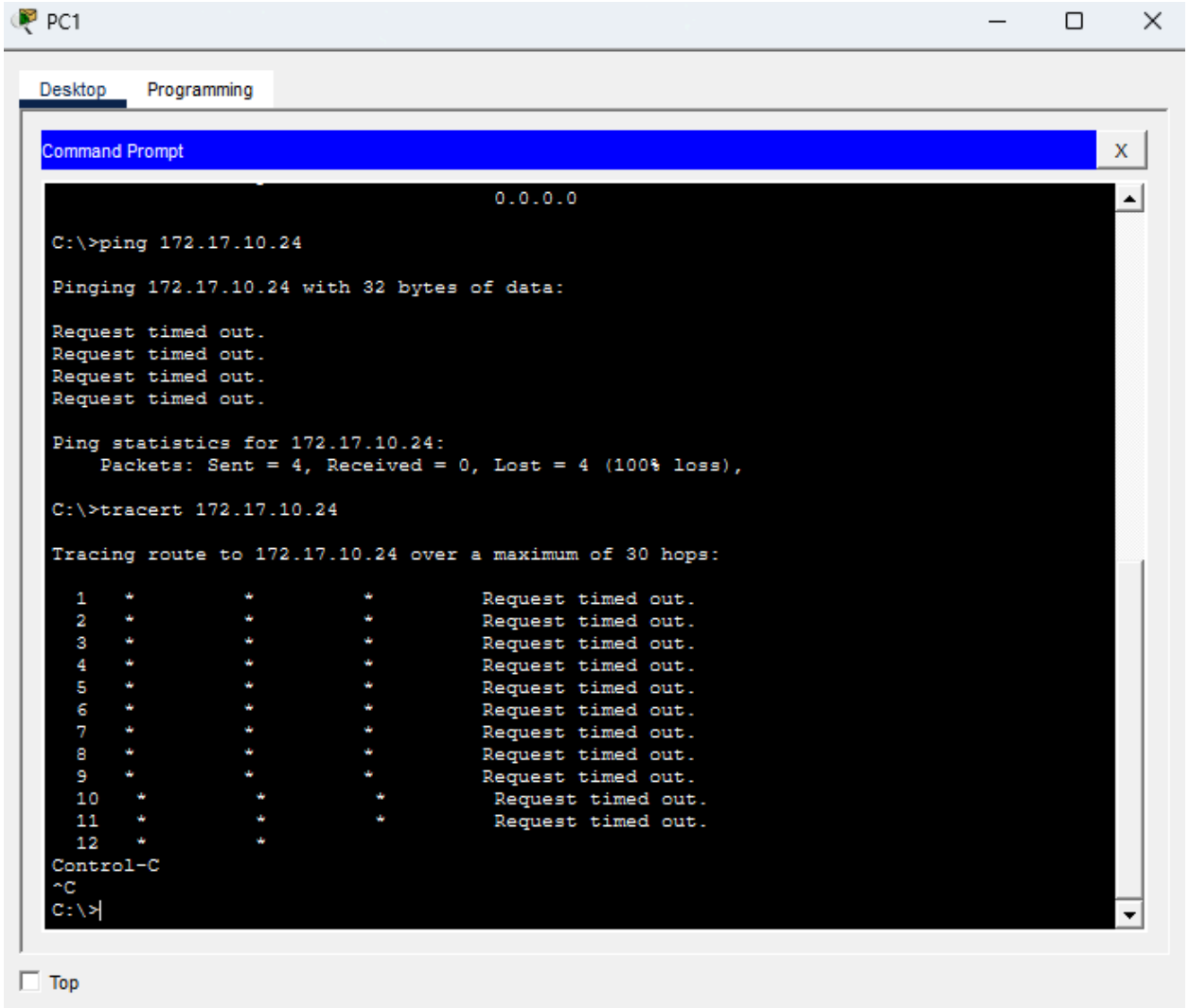


3. Master Addressing Table

Device	Interface	IP Address	Subnet Mask	Assigned VLAN
PC1	NIC	172.17.10.21	255.255.255.0	VLAN 10 (Faculty/Staff)
PC2	NIC	172.17.20.22	255.255.255.0	VLAN 20 (Students)
PC3	NIC	172.17.30.23	255.255.255.0	VLAN 30 (Guest)
PC4	NIC	172.17.10.24	255.255.255.0	VLAN 10 (Faculty/Staff)
PC5	NIC	172.17.20.25	255.255.255.0	VLAN 20 (Students)
PC6	NIC	172.17.30.26	255.255.255.0	VLAN 30 (Guest)

To quickly check that your names and interface associations match the lab design:
S2# show vlan brief

Step B: Host Network Layer Verification
On PC1 Command Prompt:
C:\> ipconfig
Expected Output: IP Address. . . . . : 172.17.10.21 Subnet Mask . . . . : 255.255.255.0
Step C: Inter-VLAN / Connectivity Testing
If you attempt to ping PC4 from PC1 immediately after this configuration:
C:\> ping 172.17.10.24
Result: Request timed out.
C:\> tracert 172.17.10.24
Result: Drops immediately at the local node.



5. Critical Analysis & Troubleshooting
Why did the pings fail?
Although PC1 and PC4 are structurally assigned to the correct logical broadcast domain (VLAN 10), the switches are linked together via standard links operating inside VLAN 1 by default.

- Looking at the provided show vlan brief output for S2, ports Gig0/1 and Gig0/2 are still explicitly stuck in VLAN 1.
- Because these inter-switch links are operating as access ports in VLAN 1, any frame tagged with VLAN 10, VLAN 20, or VLAN 30 traveling from S2 trying to cross over to S1 will be completely dropped.

How to resolve this issue?

To allow traffic from all configured VLANs to travel between S2, S1, and S3, the interconnected ports (such as $Gig0/1$ and $Gig0/2$) must be changed from Access Mode to Trunk Mode using IEEE 802.1Q encapsulation.
Remediation Command Example (On S1, S2, and S3 uplink ports):
S1

```
S1#
S1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#interface range g0/1 - 2
S1(config-if-range)#switchport mode trunk

S1(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with S2 GigabitEthernet0/1 (99).

S1(config-if-range)#switchport trunk native vlan 99
S1(config-if-range)#%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/1 on VLAN0099. Port consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/1 on VLAN0001. Port consistency restored.


%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (99), with S3 GigabitEthernet0/2 (1).

S1(config-if-range)#end
S1#
%SYS-5-CONFIG_I: Configured from console by console
```

S2

```
S2>enable
S2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S2(config)#interface range g0/1 - 2
S2(config-if-range)#switchport mode trunk

S2(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

S2(config-if-range)#switchport mode trunk
S2(config-if-range)#switchport trunk native vlan 99
S2(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (99), with S1 GigabitEthernet0/1 (1).
%SPANTREE-2-RECV_PVID_ERR: Received BPDU with inconsistent peer vlan id 1 on GigabitEthernet0/1 VLAN99.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking GigabitEthernet0/1 on VLAN0099. Inconsistent local vlan.


S2(config-if-range)#end
```

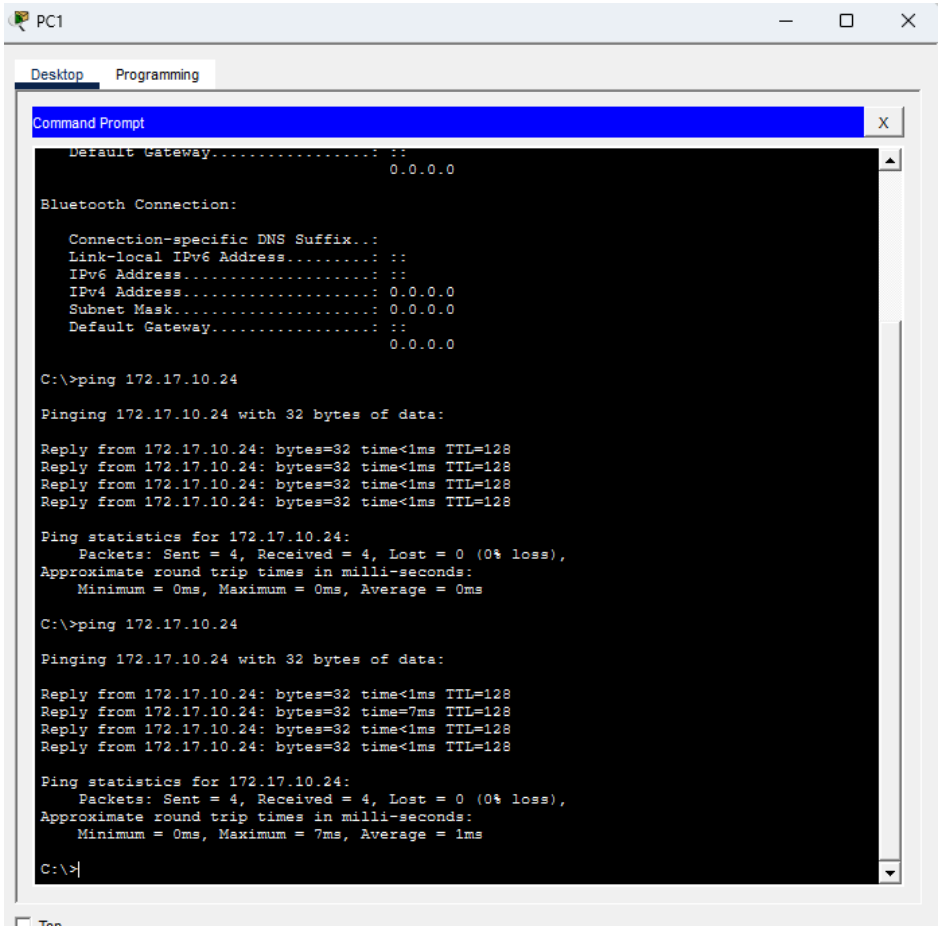
S3

```
S3>
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (1), with S1 GigabitEthernet0/2 (99).

S3>enable
S3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
S3(config)#interface range g0/1 - 2
S3(config-if-range)#switchport mode trunk
S3(config-if-range)#switchport trunk native vlan 99
S3(config-if-range)#%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/2 on VLAN0099. Port consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking GigabitEthernet0/2 on VLAN0001. Port consistency restored.


S3(config-if-range)#end
S3#
%SYS-5-CONFIG_I: Configured from console by console
```



6. Personal Reflections

● Logical vs. Physical Architecture: This lab highlights that physical proximity or simply being on the same IP subnet is no longer enough to guarantee data delivery once layer 2 micro-segmentation is introduced.

The Magic of Voice Integration: Merging voice traffic (VLAN 150) and data traffic (VLAN 10) down a single copper drop on S3 ($F0/11$) shows how enterprise networks maximize their cabling investments. Using QoS (mls qos trust cos) prevents an intensive PC download from degrading a crisp VoIP voice call.